ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

# INORGANIC CHEMISTRY

## JEE (MAIN+ADVANCED)

NURTURE COURSE




## Exercise

# Chemical Bonding

(English Medium)

# EXERCISE # O-1

## *IONIC BOND*

***(Only one option is correct)***

**1.** The compound which contains ionic as well as covalent bonds is

(A) $C_2H_4Cl_2$ (B) $CH_3I$ (C) KCN (D) $H_2O_2$

**CBO0001**

**2.** A bond formed between two like atoms cannot be

(A) ionic (B) covalent (C) coordinate (D) metallic

**CBO0002**

**3.** An ionic bond $A^+B^-$ is most likely to be formed when :

(A) the ionization energy of *A* is high and the electron gain enthalpy of *B* is low

(B) the ionization energy of *A* is low and the electron gain enthalpy of *B* is high

(C) the ionization energy of *A* and the electron gain enthalpy of *B* both are high

(D) the ionization energy of *A* and the electron gain enthalpy of *B* both are low

**CBO0003**

## *COVALENT BOND , CO-ORDINATE BOND & LEWIS STRUCTURE*

**4.** Which of the following species are hypervalent?

(1) $PCl_5$, (2) $BF_3$, (3) $XeF_2$ , (4) $CO_3^{2-}$

(A) 1, 2, 3 (B) 1, 3 (C) 3, 4 (D) 1, 2

**CBO0004**

**5.** The types of bond present in $N_2O_{5(g)}$ are

(A) only covalent (B) only ionic (C) ionic and covalent (D) covalent & coordinate

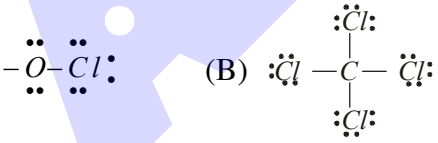
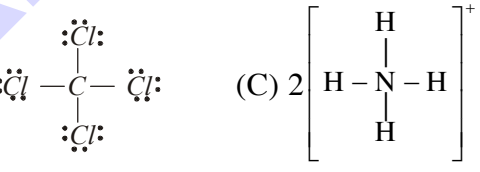
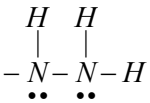
**CBO0005**

**6.** Which of the following molecule does not have coordinate bonds?

(A) $CH_3$–NC (B) CO (C) $O_3$ (D) $CO_3^{2-}$

**CBO0006**

**7.** Which of the following Lewis dot diagrams is(are) incorrect ?

(A)  (B)  (C)  (D) 

**CBO0007**

## *V.B.T., HYBRIDISATION & VSEPR THEORY*

**8.** Which of the following has a geometry different from the other three species (having the same geometry)?

(A) $BF_4^-$ (B) $SO_4^{2-}$ (C) $XeF_4$ (D) $PH_4^+$

**CBO0008**

**9.** Maximum bond energy is in :

(A) $F_2$ (B) $N_2$ (C) $O_2$ (D) equal

**CBO0009**

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**10.** The hybridisation and shape of $BrF_3$ molecule are :

(A) $sp^3d$ and bent *T* shape (B) $sp^2d^2$ and tetragonal

(C) $sp^3d$ and bent (D) none of these

**CBO0010**

**11.** The shape of methyl cation ($CH_3^+$) is likely to be:

(A) linear (B) pyramidal (C) planar (D) spherical

**CBO0011**

**12.** The structure of $XeF_2$ involves hybridization of the type :

(A) $sp^3$ (B) $sp^2$ (C) $sp^3d$ (D) $sp^3d^2$

**CBO0012**

**13.** Which of the following has been arranged in increasing order of length of the hybrid orbitals ?

(A) $sp < sp^2 < sp^3$ (B) $sp^3 < sp^2 < sp$ (C) $sp^2 < sp^3 < sp$ (D) $sp^2 < sp < sp^3$

**CBO0013**

**14.** In the context of carbon, which of the following is arranged in the correct order of electronegativity :

(A) $sp > sp^2 > sp^3$ (B) $sp^3 > sp^2 > sp$ (C) $sp^2 > sp > sp^3$ (D) $sp^3 < sp < sp^2$

**CBO0014**

**15.** When 2s–2s, 2p–2p and 2p–2s orbitals overlap, the bond strength decreases in the order :

(A) p – p > s – s > p – s (B) p – p > p – s > s – s

(C) s – s > p – p > p – s (D) s – s > p – s > p – p

**CBO0015**

**16.** Carbon atoms in $C_2(CN)_4$ are :

(A) *sp*-hybridized (B) $sp^2$-hybridized

(C) *sp*- and $sp^2$ hybridized (D) *sp*, $sp^2$ and $sp^3$- hybridized

**CBO0016**

**17.** The structure of $XeF_6$ in vapour phase is

(A) pentagonal bipyramidal (B) trigonal bipyramidal

(C) capped octahedron (D) square bipyramidal

**CBO0017**

## *BENT'S RULE AND DRAGO'S RULE*

**18.** C–H bond distance is the longest in:

(A) $C_2H_2$ (B) $C_2H_4$ (C) $C_2H_6$ (D) $C_2H_2Br_2$

**CBO0018**

**19.** The bond angle at oxygen and hybridization of oxygen in ether ($CH_3OCH_3$) is :

(A) 106°51', $sp^3$ (B) 104°31', $sp^3$ (C) 110°, $sp^3$ (D) 109°28', $sp^3$

**CBO0019**

**20.** Which of the following has been arranged in order of decreasing bond length ?

(A) P–O > Cl–O > S–O (B) P–O > S–O > Cl–O

(C) S–O > Cl–O > P–O (D) Cl–O > S–O > P–O

**CBO0020**

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**21.** Select the **CORRECT** set of statements :
As the s-character of a hybrid orbital decreases
(I) The bond angle decreases (II) The bond strength increases
(III) The bond length increases (IV) Size of orbital increases
(A) (I), (III) and (IV) (B) (II), (III) and (IV) (C) (I) and (II) (D) All are **CORRECT**

**CBO0021**

**22.** Among the following, the **CORRECT** statement is :
(A) Between $NH_3$ and $PH_3$, $NH_3$ is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional
(B) Between $NH_3$ and $PH_3$, $PH_3$ is a better electron donor because the lone pair of electrons occupies $sp^3$orbital and is more directional
(C) Between $NH_3$ and $PH_3$, $NH_3$ is a better electron donor because the lone pair of electrons occupies $sp^3$ orbital and is more directional
(D) Between $NH_3$ and $PH_3$, $PH_3$ is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

**CBO0022**

**23.** In which of the following molecule C—C bond length will be highest ?
(A) $CF_3 - CF_3$ (B) $F_2CH - CHF_2$ (C) $FCH_2 - CH_2F$ (D) $CH_3–CF_3$

**CBO0023**

**24.** In BClBr I molecule the maximum % s-character provided from the central atom is in bond :
(A) B – I (B) B – Cl (C) B – Br (D) Can not predict

**CBO0024**

**25.** Which of the following order is **CORRECT** for increasing p-character in orbital used for bonding by central atom
(A) $SiH_4 > CH_4$ (B) $H_2S > H_2O$ (C) $PH_4^+ > PH_3$ (D) $NH_3 > PH_3$

**CBO0025**

## *MOLECULE DOES NOT EXIST*

**26.** $PCl_5$ exists but $NCl_5$ does not because :
(A) Nitrogen has no vacant 2*d*-orbitals (B) $NCl_5$ is unstable
(C) Nitrogen atom is much smaller than P (D) Nitrogen is highly inert

**CBO0026**

**27.** Which of the following molecules do not exist.
(A) $BF_6^{3-}$ (B) $NCl_3$ (C) $NOF_3$ (D) $XeF_5^-$

**CBO0027**

**28.** Select non existing species
(A) $PH_3$ (B) $PH_4^+$ (C) $[PF_6]^-$ (D) None of these

**CBO0028**

**29.** Among the fluorides below, the one which does not exist is -
(A) $XeF_4$ (B) $BeF_4$ (C) $SF_4$ (D) $CF_4$

**CBO0029**

**30.** Which of the following compound exist -
(A) $XeH_2$ (B) $ClF_7$ (C) $PH_5$ (D) $ICl_3$

**CBO0030**

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## DIPOLE MOMENT & HYDROGEN BONDING

**31.** Which of the following has been arranged in order of decreasing dipole moment ?

(A) $CH_3Cl > CH_3F > CH_3Br > CH_3I$ (B) $CH_3F > CH_3Cl > CH_3Br > CH_3I$

(C) $CH_3Cl > CH_3Br > CH_3I > CH_3F$ (D) $CH_3F > CH_3Cl > CH_3I > CH_3Br$

**CBO0031**

**32.** The experimental value of the dipole moment of *HCl* is 1.03 *D*. The length of the *H–Cl* bond is 1.275 Å. The percentage of ionic character in *HCl* is nearly :

(A) 43 (B) 21 (C) 17 (D) 7

**CBO0032**

**33.** Ethanol has a higher boiling point than dimethyl ether though they have the same molecular weight. This is due to :

(A) resonance (B) coordinate bonding

(C) hydrogen bonding (D) ionic bonding

**CBO0033**

**34.** Which of the following compounds would have significant intermolecular hydrogen bonding ?

HF, $CH_3OH$, $N_2O_4$, $CH_4$

(A) HF, $N_2O_4$ (B) HF, $CH_4$, $CH_3OH$ (C) HF, $CH_3OH$ (D) $CH_3OH$, $CH_4$

**CBO0034**

**35.** For $H_2O_2$, $H_2S$, $H_2O$ and HF, the correct order of decreasing strength of hydrogen bonding is :

(A) $H_2O > HF > H_2O_2 > H_2S$ (B) $HF > H_2O_2 > H_2O > H_2S$

(C) $HF > H_2O > H_2O_2 > H_2S$ (D) $H_2O_2 > H_2O > HF > H_2S$

**CBO0035**

**36.** Which one of the following does not have intermolecular H-bonding?

(A) $H_2O$ (B) *o*-nitro phenol (C) HF (D) $CH_3COOH$

**CBO0036**

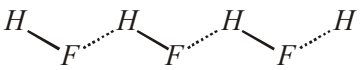
**37.** The order of strength of hydrogen bonds is:

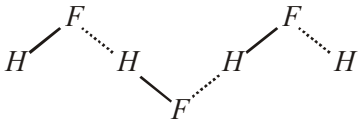
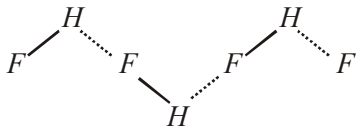
(A) NH...N > OH...O > FH...F (B) NH...N < OH...O < FH...F

(C) NH...N > OH...O < FH...F (D) NH...N < OH...O > FH...F

**CBO0037**

**38.** The *H* bond in solid *HF* can be best represented as:

(A) H – F....H – F....H – F (B) 

(C)  (D) 

**CBO0038**

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## VANDERWAAL FORCES

**39.** **Statement-1 :** The melting point of noble gases increases as its atomic mass increases.
**Statement-2 :** Instantaneous dipole induced dipole attraction increases with increase in atomic mass of noble gases.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CBO0039**

**40.** The critical temperature of water is higher than that of $O_2$ because the $H_2O$ molecule has :
(A) fewer electrons than $O_2$ (B) two ionic bonds
(C) V–shape (D) dipole moment

**CBO0040**

**41.** Which of the following boiling point order is correct -
(A) $He > T_2 > D_2$ (B) $He < T_2 < D_2$ (C) $T_2 > He > D_2$ (D) $He < D_2 < T_2$

**CBO0041**

**42.** Which is the incorrect match for the energy distance function for following interaction -
(A) Debye force : $r^{-6}$ (B) Ion-induced dipole interaction : $r^{-2}$
(C) London force : $r^{-6}$ (D) Keesom force : $r^{-3}$

**CBO0042**

**43.** Identify the incorrect order of boilng point in the following pair.
(A) $B(OH)_3 < B(OCH_3)_3$ (B) $NF_3 < N(CH_3)_3$
(C) $BF_3 < B(CH_3)_3$ (D) $C_2H_6 < C_2F_6$

**CBO0043**

## SILICATE

**44.** The number of corners or O-atoms shared per tetrahedron for pyroxene chain silicate is -
(A) 3 (B) 2 (C) 2.5 (D) 1

**CBO0044**

**45.** The mineral $Na_2Fe^{II}_3Fe^{III}_2[Si_8O_{22}](OH)_2$ (chrocidolite) is a :
(A) Pyroxene chain silicate (B) Sheet silicate
(C) Amphiboles chain silicate (D) 3D-silicate

**CBO0045**

**46.** The silicate anion in the mineral kinoite is a chain of three $SiO_4$ tetrahedra that share corners with adjacent tetrahedra. The mineral also contains $Ca^{2+}$ ions, $Cu^{2+}$ ions, and water molecules in a 1:1:1 ratio mineral is represented as :
(A) $CaCuSi_3O_{10}.H_2O$ (B) $CaCuSi_3O_{10}.2H_2O$
(C) $Ca_2Cu_2Si_3O_{10}.2H_2O$ (D) none of these

**CBO0046**

**47.** Silicate thortveitite ($Sc_2Si_2O_7$) is an example of -
(A) Ortho silicate (B) Pyro silicate (C) Cyclic silicate (D) Chain silicate

**CBO0047**

**48.** Name of the structure of silicates in which three oxygen atoms of $SiO_4^{4-}$ are shared is -
(A) Pyro silicate (B) Sheet silicate
(C) Linear chain silicate (D) Three dimensional silicate

**CBO0048**

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## *MISCELLEANEOUS*

**49.** The formal charges on the three O-atoms in $O_3$ molecule are
(A) 0, 0, 0 (B) 0, 0, –1 (C) 0, 0, +1 (D) 0, +1, –1

**CBO0049**

**50.** The types of bonds present in $CuSO_4 \cdot 5H_2O$ are
(A) electrovalent and covalent
(B) electrovalent , coordinate covalent & H-bond
(C) covalent, coordinate covalent & H-bonds
(D) electrovalent, covalent, coordinate covalent & H-bond

**CBO0050**

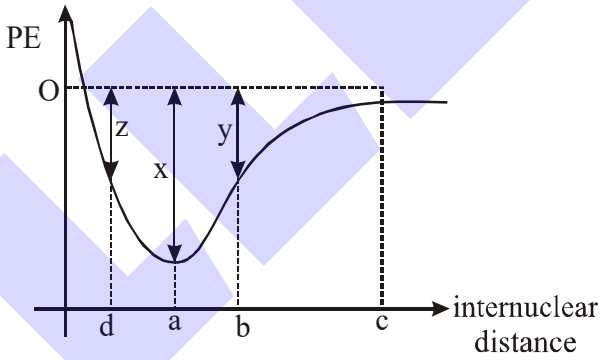
**51.** If ethylene molecule lies in X-Y plane then nodal planes of the π-bond will lie in
(A) XZ plane (B) YZ plane
(C) In a plane that bisects C–C axis (D) XY plane

**CBO0051**

**52.** Two ice cubes are pressed over each other and unite to form one cube. Which force is responsible for holding them together :
(A) van der Waal's forces (B) Covalent attraction
(C) Hydrogen bond formation (D) Dipole-dipole attraction

**CBO0052**

**53.** Consider the following graph representing potential energy and internuclear distance between two H-atoms. Select the incorrect statement-




(A) Magnitude of x gives the bond energy as well as bond dissociation energy

(B) Moving from a to d, the magnitude of x decreases to z due to increase in internuclear repulsion hence potential energy increases

(C) moving from a to b the value of x decreases to y as attractive forces becomes dominant over repulsive forces

(D) At distance c which tends to infinity, the potential energy would be only due to H-atom themselves, not because of any interaction between two atoms

**CBO0277**

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# EXERCISE # O-2

## *IONIC BOND , COVALENT BOND & COORDINATE BOND*

**1.** Which of the following compounds contain ionic, covalent and co-ordinate bonds?

(A) $NH_4Cl$ (B) KCN (C) $NaBF_4$ (D) NaOH

**CBO0053**

**2.** Which of the following statements is/are true?

(A) Covalent bonds are directional

(B) Ionic bonds are nondirectional

(C) A polar bond is formed between two atoms which have the same electronegativity value.

(D) The presence of polar bonds in a polyatomic molecule suggests that it has zero dipole moment

**CBO0054**

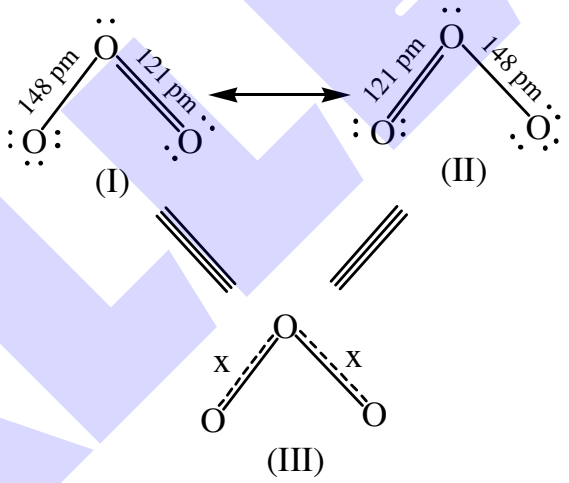
**3.** To which of the following species octet rule is not applicable :

(A) $BrF_5$ (B) $SF_6$ (C) $IF_7$ (D) CO

**CBO0055**

## *LEWIS STRUCTURE & BOND LENGTH*

**4.** $O_3$(ozone) molecule can be equally represented by the structures I and II shown below




which of the following values of x are not correct

(A) 148 pm (B) 121 pm (C) less than 121 pm (D) more than 148 pm

**CBO0056**

**5.** Select the correct statement -

(A) C–O bond length of CO molecule is higher than the C–O bond length of $CO_2$

(B) C–O bond length of CO molecule is lower than the C–O bond length of $CO_2$

(C) N–O bond length of $NO_3^-$ ion is higher than the N–O bond length of $NO_2^-$

(D) N–O bond length of $NO_3^-$ ion is lower than the N–O bond length of $NO_2^-$

**CBO0057**

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## VBT, HYBRIDISATION & VSEPR THEORY

**6.** Which of the following statement(s) is / are not correct?

(A) Hybridization is the mixing of atomic orbitals of large energy difference.

(B) $sp^2$–hybrid orbitals are formed from two *p* - atomic orbitals and one *s*- atomic orbital

(C) $sp^3d^2$–hybrid orbitals are all at 90º to one another

(D) $sp^3$–hybrid orbitals are directed towards the corners of a regular tetrahedron

**CBO0058**

**7.** Which of the following species are linear ?

(A) $ICl_2^-$ (B) $I_3^-$ (C) $N_3^-$ (D) $ClO_2^+$

**CBO0059**

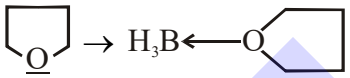
**8.** Which of the following molecules have bent shape.
Where A is central atom , B is surrounding atom and E is lone pair -

(A) $AB_2E_2$ (B) $AB_2E$ (C) $AB_3E$ (D) $AB_2$

**CBO0060**

**9.** In which of the following process(s) hybridisation of underlined atom does not change –

(A) $\underline{N}H_3 + BF_3 \rightarrow H_3N.BF_3$ (B) $\underline{Si}F_4 + 2F^{\ominus} \rightarrow [SiF_6]^{2-}$

(C) $BH_3 +$  (D) $H_3\underline{B}O_3 \xrightarrow{OH^-} [B(OH)_4]^-$

**CBO0061**

**10.** Which of the following 'd' orbital(s) participate in the hybridistaion for Xe in the cationic part of $XeF_6(s)$ :-

(A) $d_{xy}$ (B) $d_{x^2-y^2}$ (C) $d_{z^2}$ (D) $d_{y^2}$

**CBO0062**

**11.** Select the **CORRECT** statement if internuclear axis is y-axis :-

(A) $d_{xy}$ and $d_{xy}$ orbital of two atoms form π-bond

(B) $p_z$ and $p_z$ orbital of two atoms form σ-bond

(C) $d_{x^2-y^2}$ and $d_{x^2-y^2}$ orbital of two atoms form π-bond

(D) $p_y$ and $d_{zx}$ orbital of two atoms form π-bond

**CBO0063**

**12.** What is the hybridisation state of Cl in cationic part of solid $Cl_2O_6$ ?

(A) $sp^3$ (B) sp (C) $sp^2$ (D) $sp^3d$

**CBO0064**

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**13.** **INCORRECT** order of bond angle is:-

(A) $OCl_2 > SF_2$ (B) $H_2O > OF_2$ (C) $SO_4^{2-} > CF_4$ (D) $NF_3 > NH_3$

**CBO0065**

**14.** Select the **CORRECT** statement(s) ?

(A) Bond angle order : $CH_4 > CF_4$

(B) Bond length order : $d_{N-O}(NO_2^-) < d_{N-O}(NO_3^-)$

(C) Bond order of S–O : $SO_4^{2-} > SO_3^{2-}$

(D) Bond angle order : $NH_3 < PH_3$

**CBO0066**

## *MOLECULE DOES NOT EXIST*

**15.** Which of the following molecules do not exist ?

(A) $Br_2O$ (B) $SF_6$ (C) $NCl_5$ (D) $OF_4$

**CBO0067**

**16.** Which of the following do not exists ?

(A) $SH_6$ (B) $HFO_4$ (C) $SiCl_6^{-2}$ (D) $HClO_3$

**CBO0068**

**17.** Which of the following molecule exist-

(A) $SF_6$ (B) $IH_3$ (C) $SH_2$ (D) $PCl_5$

**CBO0069**

**18.** Which of the following molecules does not exist and non-existence reason is same as that of $SH_6$

(A) $OF_4$ (B) $SH_4$ (C) $PBr_6^-$ (D) $XeH_4$

**CBO0070**

## *DIPOLE MOMENT & HYDROGEN BOND*

**19.** Which of the following compounds possesses zero dipole moment?

(A) Water (B) Benzene

(C) Carbon tetrachloride (D) Boron trifluoride

**CBO0071**

**20.** Which of the following statements are correct?

(A) The crystal lattice of ice is formed by covalent as well as hydrogen bonds

(B) The density of water increases when heated from *0°C* to *4°C*

(C) Above 4° *C* the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding

(D) The density of water decreases from 0° *C* to a maximum at 4° *C*

**CBO0072**

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**21.** In which of the following chemical species hydrogen bonding takes places -
(A) $Na_2HPO_3(s)$ (B) $K_2HPO_4(s)$ (C) $KHCO_3(s)$ (D) $NaHCO_3(s)$

**CBO0073**

**22.** If $IF_x^n$, types species are planar and non-polar, then which of the following match is correct-
(where X is number of F atoms and n is charge an species)
(A) x = 2 and n = +1 (B) x = 3 and n = 0 (C) x = 2 and n = –1 (D) x = 5 and n = 0

**CBO0074**

**23.** The correct order/s of boiling point is/are -
(A) $H_2O > CH_3OH$ (B) $H_3PO_4 > Me_3PO_4$ (C) $NH_3 < H_2O$ (D) $H_2O > HF$

**CBO0075**

## VANDERWAAL FORCES

**24.** Which of the following factors are responsible for origination of vander Waals forces ?
(A) Instantaneous dipole-induced dipole interaction
(B) Dipole-induced dipole interaction
(C) Dipole-dipole interaction
(D) Size of molecule

**CBO0076**

**25.** Which of the following are true ?
(A) Vander Waals forces are responsible for the formation of molecular crystals
(B) Branching lowers the boiling points of isomeric organic compounds due to decrease in Vander Waals forces of attraction
(C) In graphite, vander Waals forces act between the carbon layers
(D) In diamond, vander Waals forces act between the carbon layers

**CBO0077**

**26.** Iodine molecules are held in solid lattice by _____.
(A) London forces (B) Dipole-dipole attraction
(C) Covalent bonds (D) Coulombic force

**CBO0078**

**27.** Which of the following inert gas liquefies easily as compare to others -
(A) Kr (B) He (C) Ne (D) Ar

**CBO0079**

## *SILICATE*

**28.** In which of the following cases the number of corner shared per tetrahedron is '2' -
(A) Pyroxene chain silicate (B) Amphibole chain silicate
(C) 5-membered cyclic silicate (D) None of these

**CBO0080**

**29.** Which of the following minerals contain three oxygen corners shared per silicate unit ?
(A) $CaMg[(SiO_3)_2]$ (B) $BaTi[Si_3O_9]$
(C) $Ca_2Mg_5[(Si_4O_{11})_2](OH)_2$ (D) $Al_2(OH)_4[Si_2O_5]$

**CBO0081**

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**30.** In Silica ($SiO_2$), each silicon atom is bonded to -

(A) Two oxygen atoms

(B) Four oxygen atoms

(C) One silicon and two oxygen atoms

(D) One silicon and four oxygen atoms

**CBO0082**

**31.** The ratio of 'Si' atom to monovalent 'O' atom in amphibole silicate is -

(A) 2 : 3 (B) 4 : 5 (C) 4 : 11 (D) 1 : 3

**CBO0083**

**32.** In which of the following silicate/s, 'Si' atoms are $sp^3$ hybrid -

(A) Pyro silicate (B) Sheet silicate (C) Pyroxene (D) Amphiboles

**CBO0084**

## *MISCELLEANEOUS & STRUCTURE*

**33.** Which of the following oxyacids of sulphur contain $S-S$ bonds ?

(A) $H_2S_2O_8$ (B) $H_2S_2O_6$ (C) $H_2S_2O_4$ (D) $H_2S_2O_5$

**CBO0085**

**34.** Molecule(s) having both polar and non polar bonds is / are

(A) $O_2F_2$ (B) $S_2Cl_2$ (C) $N_2H_4$ (D) $S_2F_{10}$

**CBO0086**

**35.** Which of the following molecule is/are planar -

(A) $CH_2Cl_2$ (B) $XeF_4$ (C) $SO_3$ (D) $NF_3$

**CBO0087**

**36.** For which of the following molecule's lewis dot structure, formal charge of central atom is same or greater than in $SO_3$ ?

(A) $SO_4^{2-}$ (B) $PO_4^{3-}$

(C) $ClO_4^-$ (D) $NO_3^-$

**CBO0278**

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## EXERCISE # S-1

***Integer Type Answer :***

**1.** In solid $\overset{+}{Na}Cl^{-}$ one $Na^{+}$ is surrounded by six $Cl^{-}$ ions find out total number of directional bonds formed by each $Na^{+}$ with $Cl^{-}$ ions.

**CBO0088**

**2.** Find out total number of $p\pi-d\pi$ bonds present in $SO_2$ molecule

**CBO0089**

**3.** Find out total number of $p\pi-p\pi$ bonds present in $SO_4^{2-}$

**CBO0090**

**4.** How many number of atomic orbitals are required, so that their mixing produces hybrid orbitals each having % of s-character nearly equal to 17%.

**CBO0091**

**5.** Calculate the maximum number of atoms lying in one plane of $PCl_3F_2$.

**CBO0092**

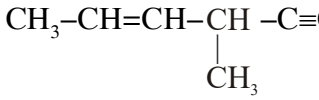
**6.** Find the total number of following molecule(s) which have all bond lengths are same.
$XeF_4$, $SF_4$, $SH_2$, $NO_3^-$, $SiF_4$, $ClF_3$, $PF_2Cl_3$, $XeO_3F_2$

**CBO0093**

**7.** A diatomic molecule has a dipole moment of 1.2 D. If bond length is 1.0 Å, what percentage of an electronic charge exists on each atom.
[Add the digits untill you are getting single digit. e.g. If the answer is 537, then 5 + 3 + 7 = 15] again 1 + 5 = 6 and 6 is to be written as answer.

**CBO0094**

**8.** Total number of $sp^3$ hybridised atom(s) in given hydrocarbon.



**CBO0095**

**9.** Find the number of lone pair on central atom of $ICl_4^-$

**CBO0096**

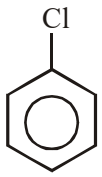
**10.** Find the number of acid(s) which are having hypo prefix in it's name from the following.
$H_3PO_4$, $H_3BO_3$, $H_3PO_3$, $H_3PO_2$, $HClO_3$, HClO, $(HNO)_2$, $H_4P_2O_6$

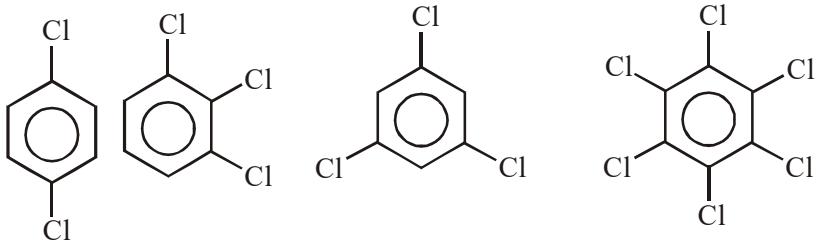
**CBO0097**

**11.** Find the number of planar molecules/ions which are cyclic –
$B_3N_3H_6$ , $C_3N_3Cl_3$ , $(BO_2^-)_3$ , $S_2O_8^{2-}$

**CBO0098**

**12.** Find the number of molecule having dipole moment less than the dipole moment of





**CBO0099**

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**13.** Find the number of molecules having intramolecular hydrogen bonding in the following :
ice, o-boric acid, o-nitrophenol, o-chlorophenol, chloral hydrate, o-salicyaldehyde

**CBO0100**

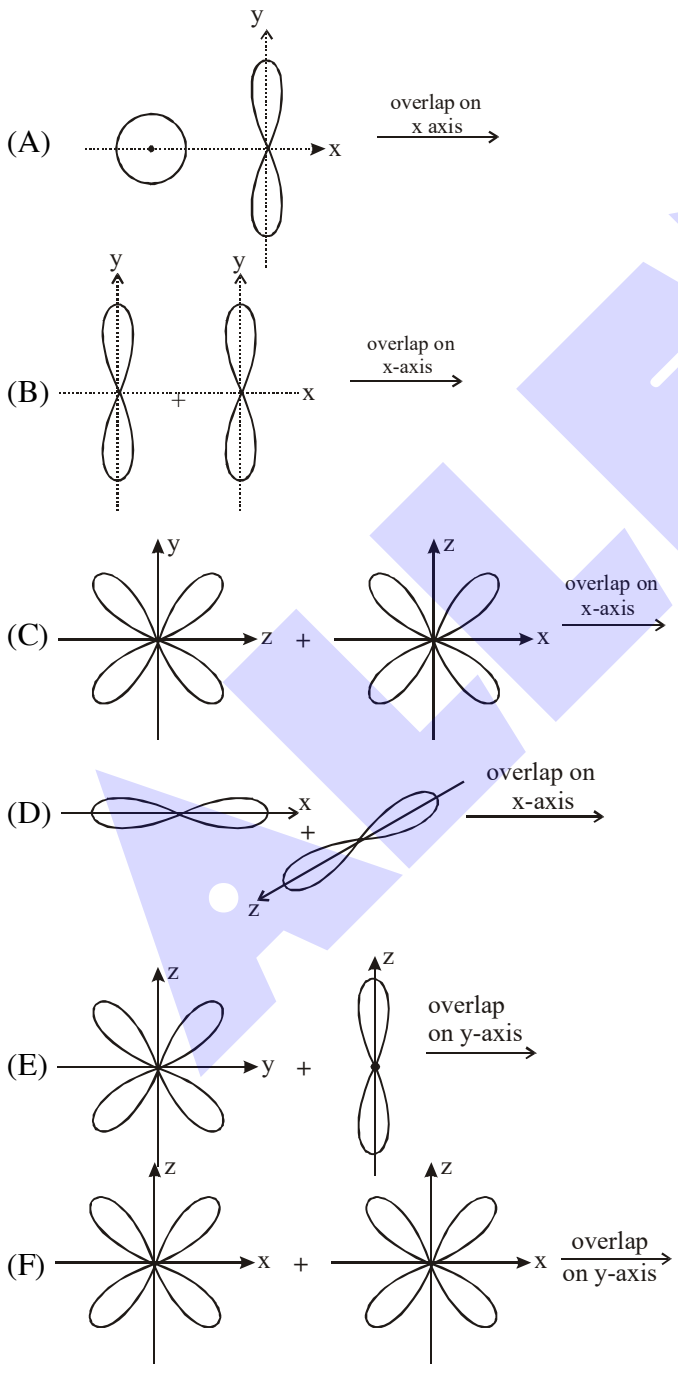
**14.** Calculate the value of "n" in $Zn_nCa_2(Si_3O_{10}).2H_2O$

**CBO0101**

**15.** In which of the following silicates structure, the number of corner/oxygen atoms shared per tetrahedron is '2'.
Pyrosilicate , pyroxene chain silicate , 2D-silicate , 3D-silicate , 4-membered cyclic silicate

**CBO0102**

**16.** How many of the following overlap results in 2-lobe 2-lobe interaction (Consider only bond forming over lapping).




**CBO0279**



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## EXERCISE # S-2

***(Comprehension) (Q.1 to Q.2)***

Bond angle and bond polarity are important parameters associated for bonding in molecule.

**1.** Which of the following bond angle order is incorrect.

(A) $NH_3 > PH_3$ (B) $BF_3 < BH_3$

(C) $O(CH_3)_2 < O(C_2H_5)_2$ (D) $CO_3^{2-} < CO_2$

**CBO0103**

**2.** Which of the following molecule is polar and non-planar

(A) $CH_2Cl_2$ (B) $PCl_5$ (C) $SO_2$ (D) $XeF_4$

**CBO0103**

***(Comprehension) (Q.3 to Q.4)***

During the formation of a covalent bond each participating atom usually acquires electronic configuration to that of nearest noble gas, and the shared electron pair remains localized between the bonded nuclei. Besides these shared electrons there are certain electron pairs which remains localized on C.A. or on substituent or both and are called non bonding or lone pair electrons.

**3.** Which of the following species does / do not follow octet rule:

(A) Hypovalent (B) Hypervalent

(C) Odd electron molecules (D) All

**CBO0104**

**4.** In $SO_3^{2-}$ the total number of lone pairs and bond pairs are respectively

(A) 18 and 4 (B) 9 and 8 (C) 9 and 4 (D) 7 and 4

**CBO0104**

***(Comprehension) (Q.5 to Q.7)***

Hybridisation is a theoretical concept, as state of hybridisation cannot be detected even by spectroscopically; unlike intermediates or transition state in various reactions. but it corrects the predictions which are based simple on overlapping of pure atomic orbitals. VSEPR theory predicts precisely shape and bond angle in a given molecule.

**5.** In which pair of molecules bond angles are not same:

(A) $CCl_4$ & $SiCl_4$ (B) $NH_4^+$ & $\overset{+}{N}F_4$ (C) $ClF_6^+$ & $SF_6$ (D) None

**CBO0105**

**6.** The molecules / ions which are planar as well as polar.

(A) $BF_3$ , $H_2O$, H F , $NH_2^-$ (B) $SnCl_2$ , $I_3^+$ , $NH_2^-$, $IF_3$

(C) $CO_3^{2-}$ , $I_3^-$, $SF_2$, $XeF_2$ (D) $NO_2^-$, $XeF_4$, $ICl_4^-$, $NH_2^+$

**CBO0105**

**7.** The correct order of energy levels of hybrid orbitals.

(A) $sp > sp^2 > sp^3$ (B) $sp < sp^2 < sp^3$ (C) $sp^2 > sp^3 > sp$ (D) $sp^3 > sp > sp^2$

**CBO0105**

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*(Comprehension) (Q.8 to Q.10)*

As one moves from sp hybridisation to $sp^3$. % of s-character in hybrid orbital decreases from 50% to 25% and p-character increases from 50% to 75% and in any hybrid orbital total % of s & p- character remains 100%. By increasing p-character the hybrid orbitals become elongated hence, their overlapping extent decreases that is results into weak bond energy also bond angle decreases.

**8.** **Statement-1 :** On decreasing s-character in hybrid orbitals, bond angle decreases.
**Statement-2 :** p-orbitals are at 90° to one another.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CBO0106**

**9.** The type of overlapping which produces bond of maximum bond energy is:
(A) $sp^3 - 1s$ (B) $sp^2 - 1s$
(C) $sp - 1s$ (D) All have same bond energy

**CBO0106**

**10.** The molecule having bond pair in which % of s character is minimum for the central atom's hybrid orbital.
(A) $SiF_4$ (B) $BCl_3$ (C) $CH_3^+$ (D) $CO_2$

**CBO0106**

*Match the Column :*

**11.** **Match the column**

| **Column-I (molecules)** | **Column-II (correct about molecule)** |
|---|---|
| (A) $\underline{P}Cl_5(\theta = 120^o), (\theta' = 90^o)$ | (P) Number of the given bond angle ($\theta$ or $\theta'$) is = One |
| (B) $\underline{N}H_3(\theta = 107^o)$ | (Q) Number of the given bond angle ($\theta$ or $\theta'$) is = Three |
| (C) $\underline{Si}H_4(\theta = 109^o\ 28')$ | (R) Number of the given bond angle ($\theta$ or $\theta'$) is = Six |
| (D) $\underline{S}O_2(\theta = 116^o)$ | (S) d-orbital(s) is/are involved in hybrdisation of central atom |
| | (T) Lone pair at underlined atom is present |

**CBO0107**

**12.**

| **Column-I** | **Column-II** |
|---|---|
| (A) $BrF_3$ | (P) Atleast one angle ≤ 90° |
| (B) $TeF_5^-$ | (Q) Central atom is $sp^3d^2$ hybridised |
| (C) $IF_7$ | (R) Non planar |
| (D) $XeF_4$ | (S) Polar |

**CBO0108**

**13.**

| **Column-I** | **Column-II** |
|---|---|
| (A) $I(CN)_2^-$ | (P) Having $p\pi-p\pi$ bond & $\mu_D = 0$ |
| (B) $CO_3^{2-}$ | (Q) Having $p\pi-d\pi$ bond & $\mu_D \neq 0$ |
| (C) $XeO_2F_2$ | (R) Planar |
| (D) $SOF_4$ | (S) Central atom is $sp^3d$ |

**CBO0109**

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**14.**

| **Column-I** | **Column-II** |
|---|---|
| (A) $H_2S_2O_5$ | (P) Central atom is $sp^3$ hybridised |
| (B) $H_6B_2O_7^{2-}$ | (Q) M–O–M i.e. oxo linkage is present |
| (C) $H_4P_2O_6$(tetrabasic) | (R) M–M linkage is present |
| (D) $H_6Si_2O_7$ | (S) Non planar |

**CBO0110**

**15.**

| **Column–I** | **Column–II** |
|---|---|
| (P) $XeF_4$ | (1) $sp^3d$ hybridised with 2 lone pair on central atom |
| (Q) $BrF_5$ | (2) $sp^3d^2$ hybridised with 2 lone pair on central atom |
| (R) $ClF_3$ | (3) $sp^3$ hybridised with 1 lone pair on central atom |
| (S) $SO_3^{2-}$ | (4) $sp^3d^2$ hybridised with 1 lone pair on central atom |

**Code :**

| | **P** | **Q** | **R** | **S** |
|---|---|---|---|---|
| (A) | 1 | 2 | 3 | 4 |
| (B) | 2 | 4 | 1 | 3 |
| (C) | 2 | 4 | 3 | 1 |
| (D) | 3 | 1 | 4 | 2 |

**CBO0111**

**16.**

| **Column–I** | **Column–II** |
|---|---|
| (P) $NH_4Cl$ | (1) diamond like structure |
| (Q) $K[HF_2]$ | (2) Benzene like structure |
| (R) $B_3N_3H_6$ | (3) Both hydrogen bond and ionic bond exist |
| (S) SiC | (4) Both covalent bond and ionic bond exist |

**Code :**

| | **P** | **Q** | **R** | **S** |
|---|---|---|---|---|
| (A) | 4 | 3 | 2 | 1 |
| (B) | 3 | 4 | 2 | 1 |
| (C) | 3 | 4 | 1 | 2 |
| (D) | 4 | 1 | 3 | 2 |

**CBO0112**

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Answer Q.17, Q.18 and Q.19 by appropriately matching the information given in the three columns of the following table.

| Column - I | Column - II | Column - III |
|---|---|---|
| (P) $d_{yz}$ | (1) $p_y$ | (i) 1 lobe – 1 lobe overlap |
| (Q) s | (2) $p_x$ | (ii) 2 lobe – 2 lobe overlap |
| (R) $d_{xz}$ | (3) $d_{yz}$ | (iii) 4 lobe – 4 lobe overlap |
| (S) $p_z$ | (4) s | (iv) Zero overlap |

**17.** Which of the following combination results in π-bond formation if internuclear axis is z-axis

(A) (P), (1), (ii) (B) (P), (4), (i) (C) (R), (4), (iv) (D) (R), (3), (ii)

**CBO0113**

**18.** Which of the following combination resultsin σ-bond formation if internuclear axis is z-axis

(A) (P), (1), (ii) (B) (Q), (3), (ii) (C) (Q), (4), (i) (D) (R), (2), (ii)

**CBO0113**

**19.** Which of the following combination results in δ-bond formation if internuclear axis is x-axis

(A) (P), (1), (i) (B) (P), (3), (iii) (C) (R), (3), (iv) (D) (P), (2), (ii)

**CBO0113**

**20.** Match the column

| **Column-I (Molecules)** | **Column-I (Characteristics)** |
|---|---|
| (A) $CH_4$ | (P) Molecule is having perfect tetrahedral shape |
| (B) $CH_2F_2$ | (Q) C-F bond has maximum p-character |
| (C) $CHF_3$ | (R) C–H bond has maximum s-character |
| (D) $CF_4$ | (S) Molecule is having maximum number of equal angles |
| | (T) Molecule has lowest bond angle |

**CBO0114**

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## EXERCISE # JEE-MAIN

**1.** The catenation tendency of C, Si and Ge is in the order Ge < Si < C. The bond energies (in kJ $mol^{-1}$) of C — C, Si —Si and Ge—Ge bonds are respectively : **[JEE-MAINS-2013 (On line)]**

(1) 348, 260, 297 (2) 348, 297, 260

(3) 297, 348, 260 (4) 260, 297, 348

**CBO0133**

**2.** In which of the following sets, all the given species are isostructural ? **[JEE-MAINS-2013 (On line)]**

(1) $BF_3$, $NF_3$, $PF_3$, $AlF_3$ (2) $PCl_3$, $AlCl_3$, $BCl_3$, $SbCl_3$

(3) $BF_4^-$, $CCl_4$, $NH_4^+$, $PCl_4^+$ (4) $CO_2$, $NO_2$, $ClO_2$, $SiO_2$

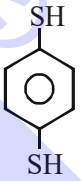
**CBO0134**

**3.** For which of the following molecule significant $\mu \neq 0$ **[JEE-M-2014]**

(1) 


(2) 


(3) 


(4) 


(1) Only (3) (2) (3) and (4) (3) Only (1) (4) (1) and (2)

**CBO0135**

**4.** Among the following oxoacids, the correct decreasing order of acid strength is :- **[JEE-M-2014]**

(1) $HClO_4 > HClO_3 > HClO_2 > HOCl$ (2) $HClO_2 > HClO_4 > HClO_3 > HOCl$

(3) $HOCl > HClO_2 > HClO_3 > HClO_4$ (4) $HClO_4 > HOCl > HClO_2 > HClO_3$

**CBO0136**

**5.** The number and type of bonds in $C_2^{2-}$ ion in $CaC_2$ are:

**[AIEEE-2005, AIEEE-2011, JEE-MAINS-2014 (On line)]**

(1) Two σ bonds and one π – bond (2) Two σ bonds and two π – bonds

(3) One σ bond and two π – bonds (4) One σ bond and one π – bond

**CBO0137**

**6.** For the compounds **[JEE-MAINS-2014]**

$CH_3Cl$, $CH_3Br$, $CH_3I$ and $CH_3F$, **(On line)**

the correct order of increasing C-halogen bond length is :

(1) $CH_3F < CH_3Br < CH_3Cl < CH_3I$ (2) $CH_3F < CH_3Cl < CH_3Br < CH_3I$

(3) $CH_3Cl < CH_3Br < CH_3F < CH_3I$ (4) $CH_3F < CH_3I < CH_3Br < CH_3Cl$

**CBO0138**

**7.** Shapes of certain interhalogen compounds are stated below. Which one of them is not correctly stated?

**[JEE-MAINS-2014]**

(1) $IF_7$ : Pentagonal bipyramid (2) $BrF_5$ : Trigonal bipyramid **(On line)**

(3) $ICl_3$ : Planar dimeric (4) $BrF_3$ : Planar T-shaped

**CBO0139**

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**8.** Which of the following molecules has two sigma(σ) and two pi(π) bonds :- **[JEE-MAINS-2014]**
(1) HCN (2) $C_2H_2Cl_2$ (3) $N_2F_2$ (4) $C_2H_4$ **(On line)**
**CBO0140**

**9.** Example of a three-dimensional silicate is : **[JEE-MAINS-2014]**
(1) Beryls (2) Zeolites (3) Feldspars (4) Ultramarines **(On-line)**
**CBO0141**

**10.** The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is :- **[JEE-M-2015]**
(1) London force (2) Hydrogen bond
(3) ion-ion interaction (4) ion-dipole interaction
**CBO0142**

**11.** Which one has the highest boiling point ? **[JEE-M-2015]**
(1) Kr (2) Xe (3) He (4) Ne
**CBO0143**

**12.** Which intermolecular force is most responsible in allowing xenon gas to liquefy?
(1) Ionic **[JEE (MAIN) ONLINE 2016]**
(2) Instantaneous dipole- induced dipole
(3) Dipole - dipole
(4) Ion - dipole
**CBO0144**

**13.** The pair in which phosphorous atoms have a formal oxidation state of + 3 is :-
(1) Pyrophosphorous and pyrophosphoric acids **[JEE-MAINS(offline)-2016]**
(2) Orthophosphorous and pyrophosphorous acids
(3) Pyrophosphorous and hypophosphoric acids
(4) Orthophosphorous and hypophosphoric acids
**CBO0145**

**14.** The group of molecules having identical shape is : **[JEE-MAINS(online)-2016]**
(1) $SF_4$, $XeF_4$, $CCl_4$ (2) $ClF_3$, $XeOF_2$, $XeF_3^+$
(3) $PCl_5$, $IF_5$, $XeO_2F_2$ (4) $BF_3$, $PCl_3$, $XeO_3$
**CBO0146**

**15.** ***Assertion :*** Among the carbon allotropes, diamond is an insulator, whereas, graphite is a good conductor of electricity. **[JEE-MAINS(online)-2016]**
***Reason :*** Hybridization of carbon in diamond and graphite are $sp^3$ and $sp^2$, respectively.
(1) Assertion is incorrect statement, but the reason is correct.
(2) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.
(3) Both assertion and reason are incorrect.
(4) Both assertion and reson are correct, but the reason is not the correct explanation for the assertion.
**CBO0147**

**16.** Aqueous solution of which salt will not contain ions with the electronic configuration $1s^22s^22p^63s^23p^6$ ? **[JEE-MAINS(online)-2016]**
(1) NaCl (2) $CaI_2$ (3) KBr (4) NaF
**CBO0148**

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**17.** The correct sequence of decreasing number of π-bonds in the structures of $H_2SO_3$, $H_2SO_4$ and $H_2S_2O_7$ is:- **[JEE-MAINS(online)-2017]**

(1) $H_2S_2O_7 > H_2SO_4 > H_2SO_3$ (2) $H_2SO_3 > H_2SO_4 > H_2S_2O_7$

(3) $H_2S_2O_7 > H_2SO_3 > H_2SO_4$ (4) $H_2SO_4 > H_2S_2O_7 > H_2SO_3$

**CBO0149**

**18.** The increasing order of the boiling points for the following compounds is :-

(I) $C_2H_5OH$ (II) $C_2H_5Cl$ **[JEE-MAINS(online)-2017]**

(III) $C_2H_5CH_3$ (IV) $C_2H_5OCH_3$

(1) (III) < (II) < (I) < (IV) (2) (II) < (III) < (IV) < (I)

(3) (IV) < (III) < (I) < (II) (4) (III) < (IV) < (II) < (I)

**CBO0150**

**19.** The number of P–OH bonds and the oxidation state of phosphorus atom in pyrophosphoric acid ($H_4P_2O_7$) respectively are:- **[JEE-MAINS(online)-2017]**

(1) five and four (2) five and five (3) four and five (4) four and four

**CBO0151**

**20.** The number of S=O and S–OH bonds present in peroxodisulphuric acid and pyrosulphuric acid respectively are : **[JEE-MAINS(online)-2017]**

(1) (2 and 4) and (2 and 4) (2) (4 and 2) and (2 and 4)

(3) (2 and 2) and (2 and 2) (4) (4 and 2) and (4 and 2)

**CBO0153**

**21.** The number of P–O bonds in $P_4O_6$ is :- **[JEE-MAINS(online)-2018]**

(1) 18 (2) 12 (3) 9 (4) 6

**CBO0154**

**22.** The decreasing order of bond angles in $BF_3$, $NH_3$, $PF_3$ and $I_3^-$ is :- **[JEE-MAINS(online)-2018]**

(1) $I_3^- > BF_3 > NH_3 > PF_3$ (2) $BF_3 > NH_3 > PF_3 > I_3^-$

(3) $I_3^- > NH_3 > PF_3 > BF_3$ (4) $BF_3 > I_3^- > PF_3 > NH_3$

**CBO0156**

**23.** Identify the pair in which the geometry of the species is T-shape and square-pyramidal, respectively :- **[JEE-MAINS(online)-2018]**

(1) $IO_3^-$ and $IO_2F_2^-$ (2) $XeOF_2$ and $XeOF_4$ (3) $ICl_2^-$ and $ICl_5$ (4) $ClF_3$ and $IO_4^-$

**CBO0157**

**24.** Which of the following is a lewis acid ? **[JEE-MAINS(online)-2018]**

(1) NaH (2) $NF_3$ (3) $PH_3$ (4) $B(CH_3)_3$

**CBO0158**

**25.** Among the oxides of nitrogen : $N_2O_3$, $N_2O_4$ and $N_2O_5$; the molecule(s) having nitrogen-nitrogen bond is/are :- **[JEE-MAINS(online)-2018]**

(1) $N_2O_3$ and $N_2O_5$ (2) $N_2O_4$ and $N_2O_5$ (3) $N_2O_3$ and $N_2O_4$ (4) Only $N_2O_5$

**CBO0159**

**26.** The incorrect geometry is represented by :- **[JEE-MAINS(online)-2018]**

(1) $BF_3$ - trigonal planar (2) $NF_3$ - trigonal planar

(3) $AsF_5$ - trigonal bipyramidal (4) $H_2O$ - bent

**CBO0160**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\1-QN, PT & CB\Eng\03_Chemical Bonding\02_Ex.

**27.** Which of the following conversions involves change in both shape and hybridisation ? **[JEE-MAINS(online)-2018]**

(1) $BF_3 \rightarrow BF_4^-$ (2) $H_2O \rightarrow H_3O^+$ (3) $CH_4 \rightarrow C_2H_6$ (4) $NH_3 \rightarrow NH_4^+$

**CBO0161**

**28.** Which of the following are Lewis acids ? **[JEE-MAINS(offline)-2018]**

(1) $AlCl_3$ and $SiCl_4$ (2) $PH_3$ and $SiCl_4$ (3) $BCl_3$ and $AlCl_3$ (4) $PH_3$ and $BCl_3$

**CBO0162**

**29.** Total number of lone pair of electrons in $I_3^-$ ion is **[JEE-MAINS(offline)-2018]**

(1) 6 (2) 9 (3) 12 (4) 3

**CBO0163**

**30.** The element that shows greater ability to form $p\pi$-$p\pi$ multiple bonds, is : **[JEE-MAINS(online)-2019]**

(1) Si (2) Ge (3) Sn (4) C

**CBO0164**

**31.** The type of hybridisation and number of lone pair(s) of electrons of Xe in $XeOF_4$, respectively, are : **[JEE-MAINS(online)-2019]**

(1) $sp^3d$ and 1 (2) $sp^3d$ and 2 (3) $sp^3d^2$ and 1 (4) $sp^3d^2$ and 2

**CBO0165**

**32.** The pair that contains two P–H bonds in each of the oxoacids is : **[JEE-MAINS(online)-2019]**

(1) $H_3PO_2$ and $H_4P_2O_5$ (2) $H_4P_2O_5$ and $H_4P_2O_6$

(3) $H_3PO_3$ and $H_3PO_2$ (4) $H_4P_2O_5$ and $H_3PO_3$

**CBO0166**

**33.** The ion that has $sp^3d^2$ hybridization for the central atom, is : **[JEE-MAINS(online)-2019]**

(1) $[ICl_2]^-$ (2) $[IF_6]^-$ (3) $[ICl_4]^-$ (4) $[BrF_2]^-$

**CBO0167**

**34.** $C_{60}$, an allotrope of carbon contains : **[JEE-MAINS(online)-2019]**

(1) 20 hexagons and 12 pentagons. (2) 12 hexagons and 20 pentagons.

(3) 18 hexagons and 14 pentagons. (4) 16 hexagons and 16 pentagons.

**CBO0168**

**35.** The oxoacid of sulphur that does not contain bond between sulphur atoms is :

(1) $H_2S_4O_6$ (2) $H_2S_2O_7$ **[JEE-MAINS(online)-2019]**

(3) $H_2S_2O_3$ (4) $H_2S_2O_4$

**CBO0169**

**36.** The number of pentagons in $C_{60}$ and trigons (triangles) in white phosphorus, respectively, are :

(1) 12 and 3 (2) 20 and 4 **[JEE-MAINS(online)-2019]**

(3) 12 and 4 (4) 20 and 3

**CBO0170**

**37.** The C–C bond length is maximum in **[JEE-MAINS(online)-2019]**

(1) graphite (2) $C_{70}$ (3) diamond (4) $C_{60}$

**CBO0171**

**38.** The dipole moments of $CCl_4$, $CHCl_3$ and $CH_4$ are in the order : **[JEE-MAINS(online)-2020]**

(1) $CH_4 = CCl_4 < CHCl_3$ (2) $CH_4 < CCl_4 < CHCl_3$

(3) $CCl_4 < CH_4 < CHCl_3$ (4) $CHCl_3 < CH_4 = CCl_4$

**CBO0172**

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**39.** The relative strength of interionic/intermolecular forces in decreasing order is : **[JEE-MAINS(online)-2020]**
(1) ion-dipole > ion-ion > dipole-dipole
(2) dipole-dipole > ion-dipole > ion-ion
(3) ion-dipole > dipole-dipole > ion-ion
(4) ion-ion > ion-dipole > dipole-dipole

**CBO0173**

**40.** The predominant intermolecular forces present in ethyl acetate, a liquid, are : **[JEE-MAINS(online)-2020]**
(1) hydrogen bonding and London dispersion
(2) Dipole-dipole and hydrogen bonding
(3) London dispersion and dipole-dipole
(4) London dispersion, dipole-dipole and hydrogen bonding

**CBO0175**

**41.** The number of bonds between sulphur and oxygen atoms in $S_2O_8^{2-}$ and the number of bonds between sulphur and sulphur atoms in rhombic sulphur, respectively, are : **[JEE-MAINS(online)-2020]**
(1) 4 and 8 (2) 4 and 6
(3) 8 and 8 (4) 8 and 6

**CBO0176**

**42.** The number of $sp^2$ hybrid orbitals in a molecule of benzene is : **[JEE-MAINS(online)-2020]**
(1) 24 (2) 6 (3) 12 (4) 18

**CBO0177**

**43.** If $AB_4$ molecule is a polar molecule, a possible geometry of $AB_4$ is : **[JEE MAINS 2020]**
(1) See-Saw (2) Tetrahedral
(3) Square planar (4) Rectangular planar

**CBO0267**

**44.** The structure of $PCl_5$ in the solid state is : **[JEE MAINS 2020]**
(1) square pyramidal (2) tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$
(3) square planar $[PCl_4]^+$ and octahedral $[PCl_6]^-$ (4) trigonal bipyramidal

**CBO0268**

**45.** The shape/structure of $[XeF_5]^-$ and $XeO_3F_2$, respectively, are : **[JEE MAINS 2020]**
(1) pentagonal planar and trigonal bipyramidal (2) trigonal bipyramidal and pentagonal planar
(3) octahedral and square pyramidal (4) trigonal bipyramidal and trigonal bipyramidal

**CBO0269**

**46.** Amongst the following, the linear species is : **[JEE MAINS 2021]**
(1) $NO_2^-$ (2) $Cl_2O$ (3) $O_3$ (4) $N_3^-$

**CBO0270**

**47.** The number of species below that have two lone pairs of electrons in their central atom is ____(Round off to the Nearest integer) **[JEE MAINS 2021]**
$SF_4$, $BF_4^-$ , $ClF_3$, $AsF_3$, $PCl_5$, $BrF_5$, $XeF_4$, $SF_6$

**CBO0271**

**48.** The correct order of bond dissociation enthalpy of halogens is : **[JEE MAINS 2021]**
(1) $Cl_2 > F_2 > Br_2 > I_2$ (2) $I_2 > Br_2 > Cl_2 > F_2$
(3) $Cl_2 > Br_2 > F_2 > I_2$ (4) $F_2 > Cl_2 > Br_2 > I_2$

**CBO0272**

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**49.** The oxidation states of 'P' in $H_4P_2O_7$, $H_4P_2O_5$ and $H_4P_2O_6$, respectively, are : **[JEE MAINS 2021]**

(1) 7, 5 and 6 (2) 5, 4 and 3 (3) 5, 3 and 4 (4) 6, 4 and 5

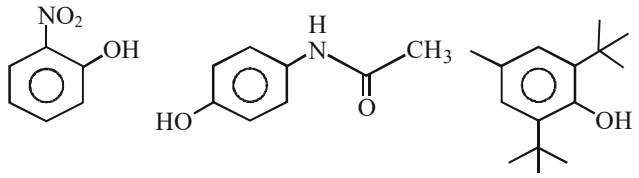
**CBO0273**

**50.** The interaction energy of London forces between two particles is proportional to $r^x$, where r is the distance between the particles. The value of x is : **[JEE MAINS 2021]**

(1) 3 (2) –3 (3) –6 (4) 6

**CBO0274**

**51.** The compound/s which will show significant intermolecular H-bonding is/are : **[JEE MAINS 2021]**




(a) (b) (c)

(1) (b) only (2) (c) only

(3) (a) and (b) only (4) (a), (b) and (c)

**CBO0275**

**52.** Amongst $BeF_2$, $BF_3$, $H_2O$, $NH_3$, $CCl_4$ and HCl, the number of molecules with non-zero net dipole moment is _____. **[JEE MAINS 2022]**

**CBO0280**

**53.** Match **List-I** with **List-II** : **[JEE MAINS 2022]**

| List-I (Molecule) | | List-II (Hybridisation ; shape) | |
|---|---|---|---|
| (A) | $XeO_3$ | (I) | $sp^3d$ ; linear |
| (B) | $XeF_2$ | (II) | $sp^3$ ; pyramidal |
| (C) | $XeOF_4$ | (III) | $sp^3d^3$ ; distorted octahedral |
| (D) | $XeF_6$ | (IV) | $sp^3d^2$ ; square pyramidal |

Choose the correct answer from the options given below:

(A) A-II, B-I, C-IV, D-III (B) A-II, B-IV, C-III, D-I

(C) A-IV, B-II, C-III, D-I (D) A-IV, B-II, C-I, D-III

**CBO0281**

**54.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** **[JEE MAINS 2022]**

**Assertion (A) :** Boron is unable to form $BF_6^{3-}$

**Reason (R) :** Size of B is very small.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(B) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(C) **(A)** is true but **(R)** is false

(D) **(A)** is false but **(R)** is true

**CBO0282**

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**55.** The number of molecule(s) or ion(s) from the following having non-planar structure is_____.

$NO_3^-$ , $H_2O_2$ , $BF_3$ , $PCl_3$ , $XeF_4$ , $SF_4$ , $XeO_3$ , $PH_4^+$ , $SO_3$ , $[Al(OH)_4]^-$ **[JEE MAINS 2022]**

**CBO0283**

**56.** Number of lone pairs of electrons in the central atom of $SCl_2$, $O_3$, $ClF_3$ and $SF_6$, respectively, are : **[JEE MAINS 2022]**

(A) 0, 1, 2 and 2 (C) 2, 1, 2 and 0 (C) 1, 2, 2 and 0 (D) 2, 1, 2 and 0

**CBO0284**

**57.** Based upon VSEPR theory, match the shape (geometry) of the molecules in List-I with the molecules in List-II and select the most appropriate option **[JEE MAINS 2022]**

| **List-I (Shape)** | **List-II (Molecules)** |
|---|---|
| (A) T-shaped | (I) $XeF_4$ |
| (B) Trigonal planar | (II) $SF_4$ |
| (C) Square planar | (III) $ClF_3$ |
| (D) See-saw | (IV) $BF_3$ |

(A) (A) - I, (B) - (II), (C) - (III), (D) - (IV)

(B) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

(C) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

(D) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

**CBO0285**

**58.** In the structure of $SF_4$, the lone pair of electrons on S is in. **[JEE MAINS 2022]**

(A) equatorial position and there are two lone pair-bond pair repulsions at 90°

(B) equatorial position and there are three lone pair-bond pair repulsions at 90°

(C) axial position and there are three lone pair - bond pair repulsion at 90°.

(D) axial position and there are two lone pair - bond pair repulsion at 90°.

**CBO0286**

**59.** Which of the following oxoacids of sulphur contains "S" in two different oxidation states?

(A) $H_2S_2O_3$ (B) $H_2S_2O_6$ (C) $H_2S_2O_7$ (D) $H_2S_2O_8$ **[JEE MAINS 2022]**

**CBO0287**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\1-QN_PT & CB\Eng\03_Chemical Bonding\02_Ex.

# EXERCISE # JEE-ADVANCED

**Integer Type**

**1.** The number of water molecule(s) directly bonded to the metal centre in $CuSO_4.5H_2O$ is **[JEE 2009]**

**CBO0178**

**2.** Based on VSEPR theory, the number of 90 degree F–Br–F angles in $BrF_5$ is **[JEE 2010]**

**CBO0179**

**3.** The difference in the oxidation numbers of the two types of sulphur atoms in $Na_2S_4O_6$ is. **[JEE 2011]**

**CBO0180**

**4.** The total number of lone-pairs of electrons in melamine is **[JEE Adv. 2013]**

**CBO0181**

**5.** The sum of the number of lone pairs of electrons on each central atom in the following species is. **[JEE 2017]**

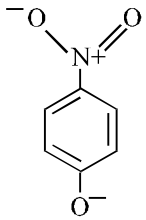
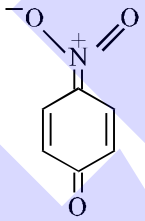
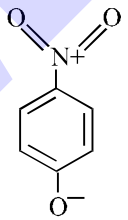
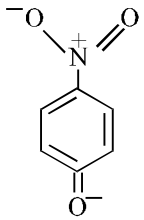
$[TeBr_6]^{2-}$, $[BrF_2]^+$ , $SNF_3$ and $[XeF_3]^-$

[Atomic number : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54]

**CBO0182**

**One or more than one correct**

**6.** The most likely representation of resonance structure of p–nitrophenoxide is:

(A)  (B)  (C)  (D) 

**CBO0183**

**7.** Specify hybridization of N and B atoms in a 1 : 1 complex of $BF_3$ and $NH_3$ **[JEE 2002]**

(A) N : tetrahedral, $sp^3$ ; B : tetrahedral, $sp^3$ (B) N : pyramidal, $sp^3$; B : pyramidal, $sp^3$

(C) N : pyramidal, $sp^3$ ; B : planar, $sp^2$ (D) N : pyramidal, $sp^3$; B : tetrahedral, $sp^3$

**CBO0184**

**8.** The nodal plane in the π-bond of ethene is located in **[JEE 2002]**

(A) the molecular plane

(B) a plane parallel to the molecular plane

(C) a plane perpendicular to the molecular plane which bisects, the carbon-carbon σ bond at right angle

(D) a plane perpendicular to the molecular plane which contains, the carbon-carbon bond.

**CBO0185**

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**9.** **Statement-1** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

***because***

**Statement-2** : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. **[JEE 2007]**

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

**CBO0186**

**10.** **Statement-1** : In water, orthoboric acid behaves as a weak monobasic acid.

***because***

**Statement-2** : In water, orthoboric acid acts as a proton donor. **[JEE 2007]**

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

**CBO0187**

**11.** The nitrogen oxide(s) that contain(s) N–N bond(s) is/are **[JEE 2009]**

(A) $N_2O$ (B) $N_2O_3$ (C) $N_2O_4$ (D) $N_2O_5$

**CBO0188**

**12.** The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is **[JEE 2010]**

**CBO0189**

**13.** The species having pyramidal shape is/are : **[JEE 2010]**

(A) $SO_3$ (B) $BrF_3$ (C) $SiO_3^{2-}$ (D) $OSF_2$

**CBO0190**

**14.** Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen-

(A) $HNO_3$, NO, $NH_4Cl$, $N_2$ (B) $HNO_3$, NO, $N_2$, $NH_4Cl$ **[JEE 2012]**

(C) $HNO_3$, $NH_4Cl$, NO, $N_2$ (D) NO, $HNO_3$, $NH_4Cl$, $N_2$

**CBO0191**

**15.** The shape of $XeO_2F_2$ molecule is : **[JEE 2012]**

(A) Trigonal bipyramidal (B) Square planar

(C) tetrahedral (D) see-saw

**CBO0192**

**16.** The compound(s) with TWO lone pairs of electrons on the central atom is(are) **[JEE 2016]**

(A) $BrF_5$ (B) $ClF_3$ (C) $XeF_4$ (D) $SF_4$

**CBO0193**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\1-QN, PT & CB\Eng\03_Chemical Bonding\02_Ex.

**17.** The crystalline form of borax has **[JEE 2016]**

(A) Tetranuclear $[B_4O_5(OH)_4]^{2-}$ unit

(B) All boron atoms in the same plane

(C) Equal number of $sp^2$ and $sp^3$ hybridized boron atoms

(D) One terminal hydroxide per boron atom

**CBO0194**

**18.** The order of the oxidation state of the phosphorus atom in $H_3PO_2$, $H_3PO_4$, $H_3PO_3$ and $H_4P_2O_6$ is **[JEE 2017]**

(A) $H_3PO_4 > H_4P_2O_6 > H_3PO_3 > H_3PO_2$

(B) $H_3PO_3 > H_3PO_2 > H_3PO_4 > H_4P_2O_6$

(C) $H_3PO_2 > H_3PO_3 > H_4P_2O_6 > H_3PO_4$

(D) $H_3PO_4 > H_3PO_2 > H_3PO_3 > H_4P_2O_6$

**CBO0195**

**19.** Based on the compounds of group 15 elements , the correct statement(s) is (are) **[JEE 2018]**

(A) $Bi_2O_5$ is more basic than $N_2O_5$

(B) $NF_3$ is more covalent than $BiF_3$

(C) $PH_3$ boils at lower temperature than $NH_3$

(D) The N–N single bond is stronger than the P–P single bond

**CBO0196**

**20.** The total number of compounds having at least one bridging oxo group among the molecules given below is_____. **[JEE 2018]**

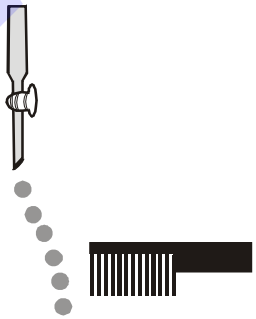
$N_2O_3$, $N_2O_5$, $P_4O_6$, $P_4O_7$, $H_4P_2O_5$, $H_5P_3O_{10}$, $H_2S_2O_3$, $H_2S_2O_5$

**CBO0197**

**21.** Consider the following compounds in the liquid form : **[JEE Adv. 2020]**

$O_2$, HF, $H_2O$, $NH_3$, $H_2O_2$, $CCl_4$, $CHCl_3$, $C_6H_6$, $C_6H_5Cl$.

When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure?



**CBO0276**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\1-GN, PT & CB\Eng\03_Chemical Bonding\02_Ex

# ANSWER KEY

## EXERCISE # O-1

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | C | A | B | B | D | D | A | C | B | A |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | C | C | A | A | B | C | C | C | C | B |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | A | C | C | A | B | A | A | D | B | D |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | A | C | C | C | C | B | B | C | A | D |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | D | B | A | B | C | C | B | B | D | D |
| Que. | 51 | 52 | 53 | | | | | | | |
| Ans. | D | C | C | | | | | | | |

## EXERCISE # O-2

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A,C | A,B | A,B,C | A,B,C,D | B,C | A,C | A,B,C | A,B | A,C | B,C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A | C | C,D | B,C | C,D | A,B,C | A,C,D | B,D | B,C,D | A,B,C |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | B,C,D | C | A,B,C,D | A,B,C | A,B,C | A | A | A,C | D | B |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | | | | |
| Ans. | A | A,B,C,D | B,C,D | A,B,C,D | B,C | A,B,C | | | | |

## EXERCISE # S-1

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 0 | 1 | 0 | 6 | 4 | 4 | 7 (25%) | 3 | 2 | 4 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | | | | |
| Ans. | 3 | 3 | 4 | 2 | 2 | 2.00 | | | | |

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\1-QN, PT & CB\Eng\03_Chemical Bonding\02_Ex.

## EXERCISE # S-2

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | A | D | C | D | B | B | B | C | A |
| Que. | 11 | | | | 12 | | | | | |
| Ans. | (A)→Q,R,S;(B)→Q,T;(C)→R;(D)→P,T | | | | (A)→P,S (B)→P,Q,R,S (C)→P,R (D)→P,Q | | | | | |
| Que. | 13 | | | | 14 | | | | | |
| Ans. | (A) → P,R,S (B) → P,R (C) → Q,S (D) → Q,S | | | | (A) → P,R,S (B) → P,Q,S (C) → P,R,S (D) → P,Q,S | | | | | |
| Que. | 15 | 16 | 17 | 18 | 19 | 20 | | | | |
| Ans. | B | A | A | C | B | (A) → P,S (B) → Q,T (C) → R (D) → P,S | | | | |

## EXERCISE # JEE-MAIN

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 2 | 3 | 2 | 1 | 3 | 2 | 2 | 1 | 2,3,4 | 2 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 2 | 2 | 2 | 2 | 4 | 4 | 1 | 4 | 3 | 4 |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | 2 | 1 | 2 | 4 | 3 | 2 | 1 | 3 | 2 | 4 |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | 3 | 1 | 3 | 1 | 2 | 3 | 3 | 1 | 4 | 3 |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | 3 | 4 | 1 | 2 | 1 | 4 | 2 | 3 | 3 | 3 |
| Que. | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | |
| Ans. | 1 | 3 | A | B | 6 | B | B | A | A | |

## EXERCISE # JEE-ADVANCED

1. Ans. 4
2. Ans. 0
3. Ans. 5
4. Ans. 6
5. Ans. 6

| Que. | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|
| Ans. | A | A | A | D | C | A,B,C or A,C | 3 | D | B |
| Que. | 15 | 16 | 17 | 18 | 19 | 20 | 21 | | |
| Ans. | D | B,C | A,C,D | A | A,B,C | 5 or 6 | 6 | | |

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